NISR
NATIONAL INSTITUTE OF STATISTICS OF RWANDA

NISR 2026 Big Data Hackathon Submission Webinar

NISR Data Science Team

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Which Platform Should I Use?

My project is built with	Use
Streamlit (Python)	**Streamlit Community Cloud**
Flask, Django, FastAPI, Node.js	**Render or PythonAnywhere**
R Shiny or Shiny for Python	**shinyapps.io**
HTML, CSS, JavaScript, React, Vue, Next.js	**Vercel, Netlify or GitHub Pages**

Streamlit Community Cloud

- **Best for:** dashboards and ML demos
- **Frameworks:** Streamlit only (Python)
- **You need:** a public GitHub repo with app.py and requirements.txt

Free tier limits

- About 2.7 GB of memory
- App sleeps after 12 hours with no visitors
- Anyone can wake it with one click

Deploy on Streamlit Cloud

1. Push app.py and requirements.txt to a public GitHub repo
2. Sign in to Streamlit Community Cloud with GitHub
3. Pick your repo, branch and app.py
4. Click Deploy and copy the app link into the form

Documentation

- Deploy your app (streamlit guide)

Render

- **Best for:** full web apps with a backend
- **Frameworks:** Flask, Django, FastAPI, Node.js and more
- **You need:** your code on GitHub

Free tier limits

- Spins down after 15 minutes without traffic
- Takes about a minute to wake up
- 750 free hours a month
- Local files are lost when it spins down

Deploy on Render

1. Push your code to GitHub
2. Create a new Web Service on Render and connect the repo
3. Set the build and start commands for your framework
4. Deploy and open your onrender.com link

Documentation

- Your first deploy
- Deploy Flask
- Deploy Django
- Deploy FastAPI

PythonAnywhere

- **Best for:** small Python web apps
- **Frameworks:** Flask and Django (Python)
- **You need:** a free account, no GitHub required

Free tier limits

- 1 web app
- 512 MiB of disk space
- 100 CPU seconds
- App expires after 1 month, so renew it

Deploy on PythonAnywhere

1. Create a free PythonAnywhere account
2. Upload your code or clone it from GitHub
3. Add a new web app from the Web tab
4. Reload the app and open your pythonanywhere.com link

Documentation

- Free account features
- Setting up Flask

shinyapps.io

- **Best for:** R Shiny dashboards and Shiny for Python
- **Frameworks:** Shiny for R, Shiny for Python (3.7 to 3.12)
- **You need:** a free shinyapps.io account

Free tier limits

- Up to 5 apps
- 25 active hours a month
- After that the app shows “not available” until next month

Deploy on shinyapps.io

1. Install the rsconnect package
2. Connect your account using your token
3. Run deployApp() from your app folder
4. Open your shinyapps.io link and test it

Documentation

- Getting started guide

Vercel

- **Best for:** front-end sites and dashboards
- **Frameworks:** Next.js, React, Vue and other JavaScript frameworks, plain HTML
- **How:** import your GitHub repo and deploy

Free tier limits

- Hobby plan: personal, non-commercial use only
- 100 GB of data transfer a month

Docs

- Hobby plan
- Builds and deploys

Netlify

- **Best for:** static websites and front-end projects
- **Frameworks:** HTML, React, Vue and other static site builders
- **How:** deploy straight from your repository

Free tier limits

- 300 credits a month
- 15 credits per production deploy
- Deploy previews are free

Docs

Deploy from a repository

GitHub Pages

- **Best for:** simple static sites hosted from your repository
- **Frameworks:** HTML, CSS, JavaScript or a static site generator
- **How:** turn on Pages in your repository settings
- **NB:** it cannot run Python or any server code

Free tier limits

- Published site up to 1 GB
- Soft bandwidth limit of 100 GB a month
- No commercial sites

Docs

Pages quickstart

Before You Submit

If your project is too big

- Use a smaller or compressed model
- Pre-compute predictions and ship the results file
- Load the model only when it is needed
- Deploy a lighter version of your app

Final checks

- Test your link in a private window
- Open the link shortly before the deadline so it is awake when we check

How to Submit

- Submit your work on provided submission link: https://statistics.gov.rw/form/2026-hackathon-submission-form
- Submissions are expected by the **30 October 2026** in a public GitHub link and a deployed app link and documentation
- **deployed app link should:**

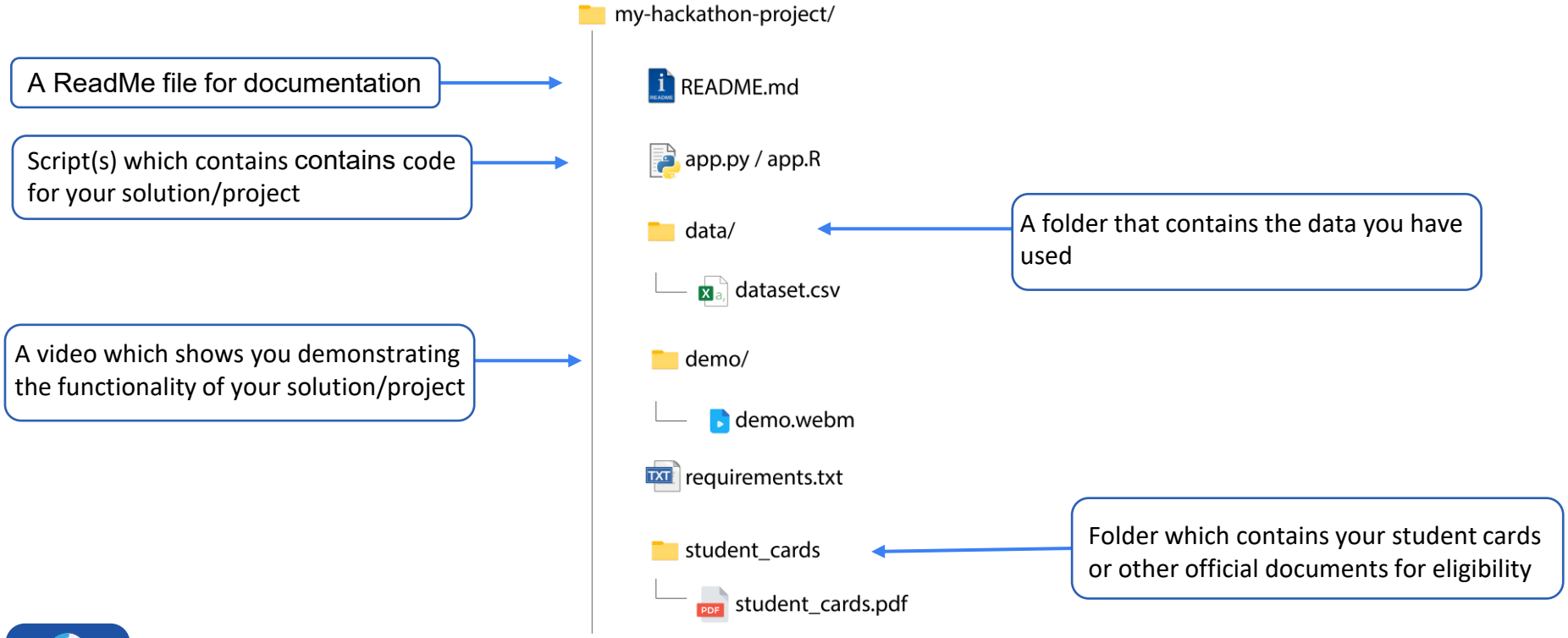
1. Be publicly accessible.
2. If login is required, provide sample credentials so users can log in easily.

- **GitHub repository should follow the criteria on next slide (at least 5 files)**:

Your GitHub repository should have at least 5 files:

1. A ReadMe file explaining what data your dashboard/application uses, what information it is trying to communicate, and how users can interact with what you're displaying.
2. A script(s) which contains the code for your solution/project which should work on any machine.
3. A video file (likely to be .webm or .gif) which shows you demonstrating the user interface and functionality of your solution/project, Do not use your mobile phone to record the screen unless you are recording a mobile app or mobile-based solution that you developed. (3 minutes maximum)
4. A file/folder containing the data you are using and a link to the data. If the data file(s) are too large, provide only the link and note this in your README file.
5. A file/folder containing your student cards or other official documents confirming your eligibility (for both members)

Sample Repository Structure



Demo: Deploying simple streamlit app

- ✓ Develop streamlit project
- ✓ Put it on GitHub
- ✓ Deploy it on streamlit community cloud
- ✓ Get the link

Demo: Submitting

- ✓ Demo on how to submit